

Who Leaves Teaching, and Why?

Teacher Attrition in the United States, 2005–2025

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Abstract

No data in the United States follow a teacher out of the classroom, so the national statistics count how many leave but not where they go, or at what price. This paper characterizes teacher turnover between 2005 and 2025 by linking Current Population Survey interviews taken twelve months apart, for every full-time classroom teacher the survey observes, following each leaver into her next occupation or out of the labor force. Each year 13 percent of teachers stop teaching for good, a rate that rose by a third over two decades entirely through moves into other jobs. Most of these moves are short and few of them pay. Two thirds of leavers are employed a year later, six in ten of those in education or care occupations, and the median leaver who moves out of education earns no more than she earned teaching while bearing twice the earnings risk of a stayer. Exit concentrates among private-school and lower-paid teachers and at both career extremes rather than in any demographic group. Estimated on seven comparable professions, the identical design shows that overall turnover in teaching is unexceptional, while a new baby raises a teacher's probability of leaving the profession by 6 percentage points, an effect absent in every comparison profession and among fathers.

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1. Introduction

Teachers are the main school input in the production of student achievement (Rivkin, Hanushek and Kain, 2005; Chetty, Friedman and Rockoff, 2014), and replacing one is costly and disruptive for the students left behind (Ingersoll, 2001; Ronfeldt, Loeb and Wyckoff, 2013). Which teachers leave, and why, is therefore a first-order question for education policy, sharpened by the fear of a post-pandemic exodus. Answering it well requires knowing not only how many teachers leave but where they go, since a teacher who moves to a better-paying school, one who switches to a job outside education, and one who stops working altogether pose different problems and call for different policies. On the second question the national evidence is nearly silent.

Three literatures measure teacher attrition in the United States, and each sees a teacher leave without seeing what she does next. The federal Teacher Follow-up Survey re-interviews a probability sample of the prior year's teachers and reports an annual leaver rate near 8 percent, the figure that anchors the policy debate.² That survey, however, runs only occasionally, its latest wave the first in nearly a decade, records each leaver's destination once and in broad categories, and cannot separate a permanent exit from a temporary pause. State administrative records support the most credible causal work on teacher mobility, but each covers a single state and loses sight of a teacher the moment she leaves its public payroll, so it observes that she left, not where she went.³ The existing national estimates come from the retrospective March supplement of the Current Population Survey, which asks respondents the occupation of their main job one year earlier; applied to teachers it yields annual rates of 6 to 8 percent (Harris and Adams, 2007; Aldeman and Yi, 2025), but a single recall question understates job-to-job movement and cannot observe returns to the classroom.

I build the missing source from the same survey. The CPS re-interviews every address twelve months apart, and by linking those interviews I follow every full-time classroom teacher it observes between 2005 and 2025, from elementary through secondary school, public or private, into her next occupation or out of the labor force, 148,390 linked observations of 47,633 distinct teachers. The design answers a plain question, the share of the people teaching school this year who will no longer be teaching school next year, and what they do instead. Because one in six apparent leavers is back in a classroom within three months, my main measure counts as a leaver only a teacher who is out of teaching at the twelve-month mark and in every re-interview after it.

Three findings organize the paper.

The first is about how much. Every year 13.1 percent of teachers stop teaching without an observed return, up from 11.5 percent in 2005–2010 to about 15 in 2022–2024. The rise is made entirely of moves into other occupations, which grew from 7.3 to 10.7 percent of teachers per year while exits into non-participation stayed flat, so the classroom is losing people to other jobs, not to inactivity, and 30 percent of leavers never leave education.

The second is about who departs, and through which door. Exit concentrates among the loosely attached, since private schools, lower pay and both career extremes predict leaving while demographics barely

²The rate is 8.0 percent for public and 12.0 percent for private school teachers in 2021–22 (NCES, 2024). The policy syntheses of teacher turnover build on the same survey and its predecessor, the Schools and Staffing Survey (Tan, Wei, Carver-Thomas and García, 2026).

³Among many others, Hanushek, Kain and Rivkin (2004) study Texas, Biasi (2021) studies Wisconsin, and the post-pandemic studies of Bacher-Hicks, Chi and Orellana (2023), Bastian and Fuller (2023), Goldhaber and Theobald (2023) and Camp, Zamarro and McGee (2025) examine Massachusetts, North Carolina, Washington and Arkansas. The meta-analysis of Nguyen, Pham, Crouch and Springer (2020) synthesizes this literature.

matter, and the doors are ordered. Better options pull teachers into other occupations, which is where all the growth is, while family events push mothers, and only mothers, out of the labor force.

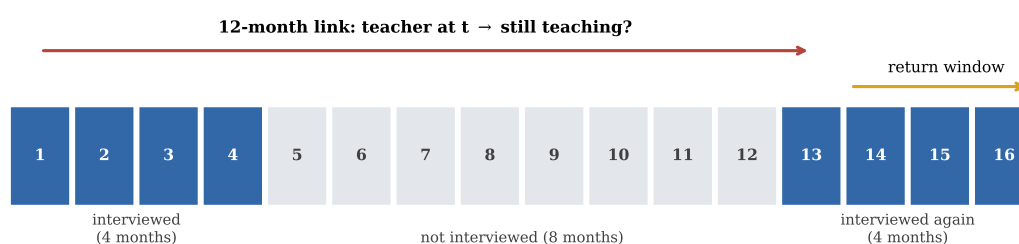
The third is about what is peculiar to teaching, and it is not the amount, since under the identical design teachers leave teaching more slowly than nurses leave nursing or pharmacists leave pharmacy. It is the motherhood door. A new baby raises a teacher's probability of leaving the profession by 6 percentage points, the national counterpart of the cohort evidence in [Stinebrickner \(2002\)](#), while the same event moves no other profession I can follow and fathers are unaffected everywhere.

None of the estimates is causal; they are the descriptive national infrastructure that the causal literature, overwhelmingly built on single states, currently lacks. Section 2 describes the data, Section 3 who teaches, Section 4 how many leave, Section 5 where they go, Section 6 who leaves and why, Section 7 places teaching among the professions, and Section 8 concludes with policy. Appendix A sets my series beside every published benchmark; Appendix B holds the full estimation output.

2. Data and definitions

I use the basic monthly microdata of the Current Population Survey, the household survey behind the official employment statistics of the United States, assembled for every month between January 2005 and December 2025.⁴ Each household is interviewed during four consecutive months, rests for eight, and is interviewed again during four more (Figure 1), so an adult interviewed in year t is re-interviewed in the same calendar month of $t+1$. I link the two interviews through household and person identifiers, validating every match on sex, race and a plausible age progression ([Madrian and Lefgren, 2000](#)); linking the public files this way is how the Bureau of Labor Statistics builds its official labor force flows and how IPUMS builds its longitudinal keys ([Rivera Drew, Flood and Warren, 2014](#)). Teachers are identified by the occupation of their main job, classroom teaching from elementary through secondary school including special education,⁵ working full time and holding at least a bachelor's degree.

Figure 1: The CPS follows each person for at most 16 months.



Notes: Interview calendar of one rotation group; numbers are months since the first interview. The 12-month link compares an interview from the first block with the interview held exactly one year later, and the return window uses the remaining interviews of the second block.

The object I measure is deliberately plain. Of everyone teaching school in a given month, public or private,

⁴The files come from the U.S. Census Bureau for 2024–2025 and from the NBER mirror for 2005–2023. October 2025 was never released because the federal government shutdown suspended that month's data collection.

⁵Census occupation codes 2310, 2320 and 2330, identical in the 2002, 2010 and 2018 classifications used over the period. The neighboring code merges preschool with kindergarten teachers and is excluded from the universe, although a move into it still counts as teaching.

full or part time, I ask who is no longer teaching school twelve months later, and what she is doing instead. Temporary exits such as parental leave would inflate that count, and indeed 15.6 percent of 12-month leavers are teaching again within three months, so my main outcome is the non-returning leaver, a teacher out of teaching at $t+12$ and in every later re-interview, computed on the 110,130 baseline observations, from 45,109 distinct teachers, whose rotation allows at least one re-interview. A person contributes up to four linked baselines and standard errors are clustered by household throughout, which nests the person. Appendix A traces the population funnel, shows that linkage does not select against teachers, reconciles my rates with every published benchmark, and states the caveats that make my levels upper bounds on permanent exit.

3. Who teaches

Before asking who leaves, it helps to know who teaches. Table 1 draws that portrait on every person I observe in a school-teaching occupation over the two decades, 116,649 individuals who represent about 5.2 million teachers in an average month, split by sex and set against the rest of the college-educated workforce; the portrait needs no degree, hours or level restriction and no successful link, which is why its universe is wider than the 47,633 teachers of the attrition panel. The typical teacher is a married woman in her early forties with children at home. Teaching is overwhelmingly female, at 78 against 47 percent, disproportionately white, and markedly less open to Asian and foreign-born graduates. Credentials cluster in the middle, with 45 percent holding a master's degree, ten points above other graduates, while doctorates remain rare. Almost three quarters work for the public sector against 15 percent elsewhere, 45 percent are union members against 9, and the price of this stability is pay, since the median teacher earns \$1,154 a week, roughly a quarter less than comparable college-educated workers.

The profile has been moving, and Figure 2 traces that movement year by year. The workforce is aging, with the share under 35 down from 32 to 29 percent, becoming markedly more Hispanic, from 6 to 13 percent, and steadily more credentialed, with master's degrees rising from 40 to 47 percent, while two anchors of the traditional teaching career erode at the same time, public-sector employment falling from 75 to 69 percent and union membership from 49 to 42. The gender mix, in contrast, has not moved at all. One of the eroding anchors matters directly for what follows, because the public sector is precisely the trait that Section 6 will show holds teachers in place, so the workforce has been drifting toward the configuration that leaves.

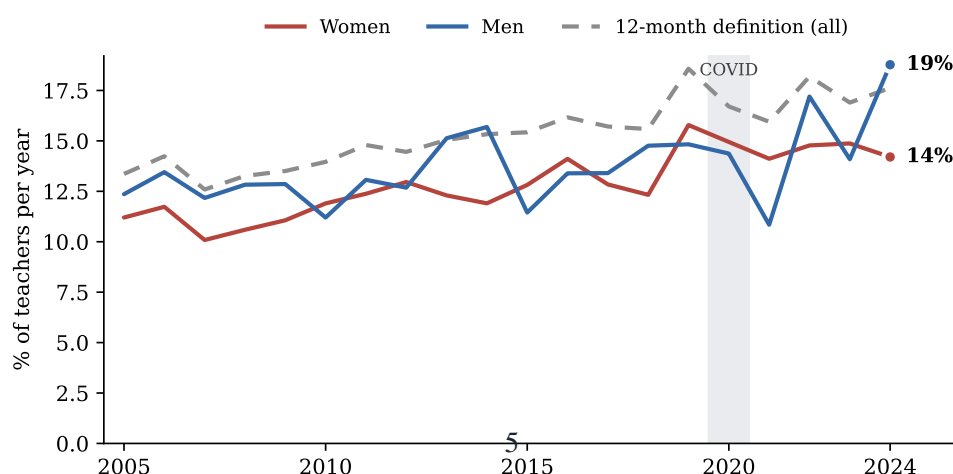
4. How many leave

Attrition is high, and it has been rising for two decades. Net of returns, it averaged 11.5 percent per year in 2005–2010 and about 15 percent in 2022–2024, with its highest values in 2019 and again from 2022 onward (Figure 3). Gender gaps are small and unstable throughout, and in the most recent year men, at 18.8 percent, exceeded women, at 14.2. This level is higher than the 8 percent of the official follow-up survey and the 6 to 8 percent of the retrospective estimates, because a panel that links two interviews sees short job-to-job moves that a single follow-up or a one-year recall question does not; Appendix A sets my rate beside every published estimate and shows that applying the recall instrument to my own data reproduces the lower figures.

Table 1: Portrait of the American teaching workforce, 2005–2025.

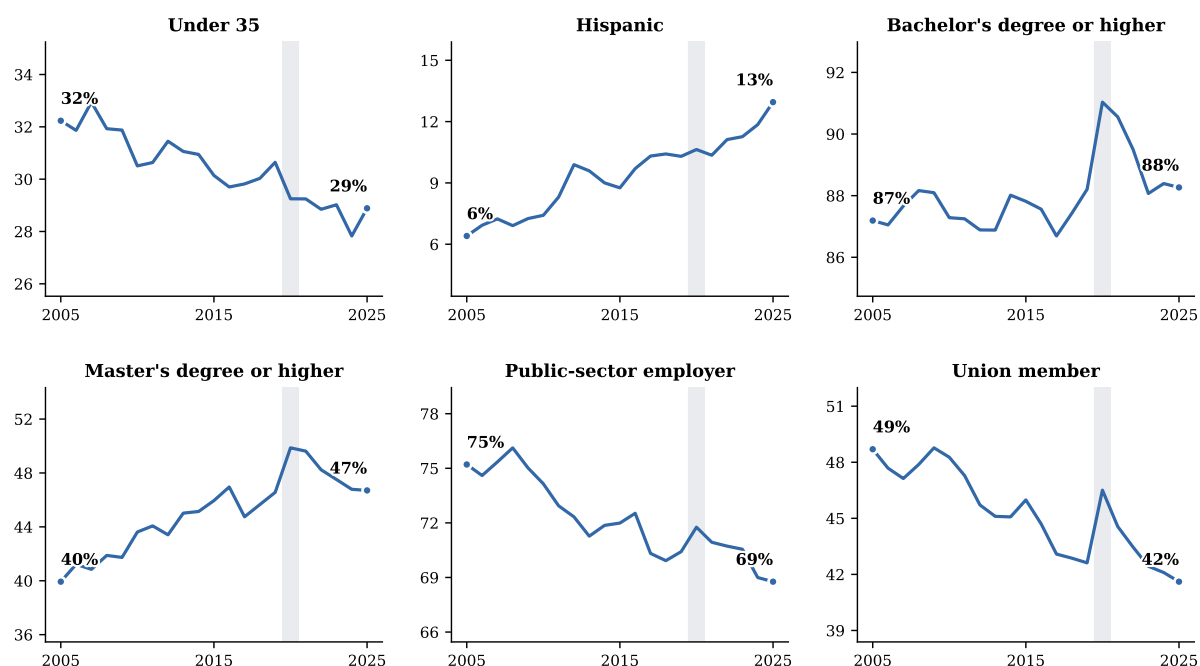
	School teachers			Other college- educated workers	Difference
	Women	Men	All		
Panel A. Demographics					
Age, years	42.6	43.1	42.7	43.8	-1.11***
Female			78.2%	47.3%	30.9***
White	84.8%	85.3%	84.9%	78.5%	6.4***
Black	10.3%	9.8%	10.2%	9.0%	1.2***
Asian	2.8%	2.6%	2.8%	10.3%	-7.5***
Hispanic	9.4%	9.3%	9.4%	8.3%	1.0***
Non-citizen	2.2%	2.3%	2.3%	7.2%	-4.9***
Panel B. Family					
Married	65.8%	69.1%	66.6%	63.8%	2.7***
Number of own children (<18) at home	0.8	0.8	0.8	0.6	0.15***
Child under 6 at home	16.7%	19.9%	17.4%	15.6%	1.8***
Panel C. Education					
Bachelor's degree or higher	86.2%	94.4%	88.0%	100.0%	-12.0***
Master's degree or higher	43.5%	50.8%	45.1%	34.5%	10.6***
Professional degree or doctorate	2.1%	3.9%	2.5%	10.5%	-8.0***
Panel D. The job					
Public-sector employer	70.5%	78.0%	72.1%	15.4%	56.8***
Union member ^a	43.9%	50.2%	45.3%	8.7%	36.6***
Usual weekly hours	39.5	41.4	39.9	40.4	-0.49***
Part-time (<35 h/week)	11.7%	6.8%	10.6%	12.3%	-1.7***
Holds more than one job	6.4%	11.3%	7.4%	6.1%	1.3***
Panel E. Pay and income					
Weekly earnings, median ^b	\$1,115	\$1,308	\$1,154	\$1,500	-\$346***
Family income \$75k+ ^b	81.1%	84.7%	81.9%	85.5%	-3.6***
Persons	92,044	24,812	116,649	963,782	
Population represented, millions ^c	4.1	1.1	5.2	49.6	

Notes: Weighted means over every employed person observed in a school-teaching occupation, with no degree restriction, pooling all 251 survey months; the comparison group is employed persons outside teaching holding at least a bachelor's degree. Sample sizes are unique persons, and a small number of persons with inconsistent sex codes across interviews appear in both sex columns. The last column reports the difference between all teachers and the comparison group, in percentage points for shares, and the significance of a two-sample test with standard errors clustered by household (** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$).^a Union membership and weekly earnings are only asked in the outgoing rotations, months in sample 4 and 8. ^b Measured over 2021–2025 only, to avoid mixing twenty years of price levels. ^c Population represented by the group in an average month. Source: author's calculations from CPS basic monthly microdata, 2005–2025.

Figure 3: Attrition net of returns, by year and gender.

Notes: Weighted share of teachers not teaching twelve months later and in no later re-interview, by baseline year and sex. The return window is capped by the survey design at three monthly interviews after the twelve-month link. The shaded band marks

Figure 2: The changing profile of the teaching workforce.

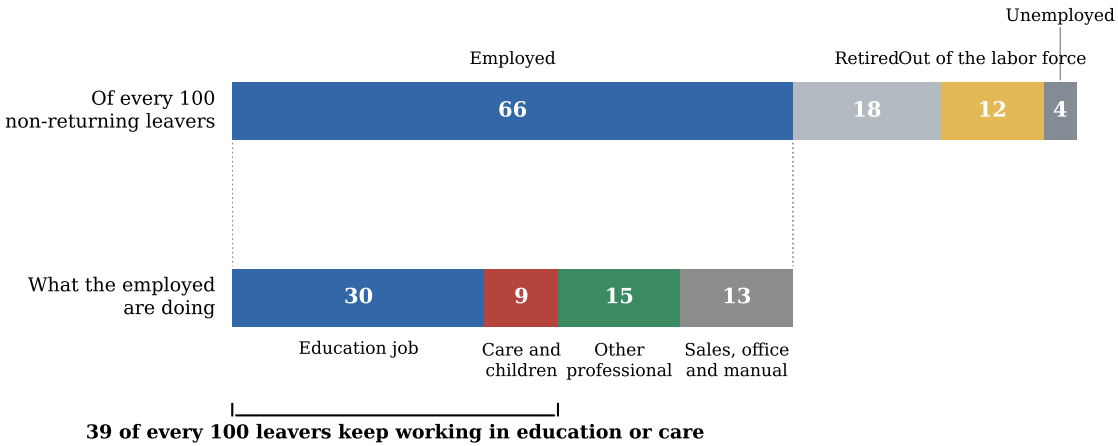


Notes: Weighted annual means over all employed persons observed in a school-teaching occupation, with no degree restriction. Union membership is only asked in the outgoing rotations. The shaded band marks 2020. Source: author's calculations from CPS basic monthly microdata, 2005–2025.

5. Where they go

Leavers do not go far, and they rarely go out of work. Figure 4 follows every 100 non-returning leavers. A year later 66 are employed, 18 have retired at a mean age of 61, 12 are out of the labor force for other reasons and only 4 are unemployed. Among the employed, education itself is the dominant destination, 30 of the original 100, whether school administration, tutoring and instruction, teaching assistance, postsecondary teaching or librarianship, and another 9 work in care occupations such as counseling, childcare or nursing, so 39 of every 100 leavers keep working in education or care and only 27 land in work unrelated to either (Appendix Figure A1 reports the detailed occupations). The gender contrast is sharp exactly where the model of Section 6 will point, since 14 percent of women who leave are out of the labor force for reasons other than retirement or disability, at ages consistent with family care, against 6 percent of men.

Figure 4: Of every 100 non-returning leavers.

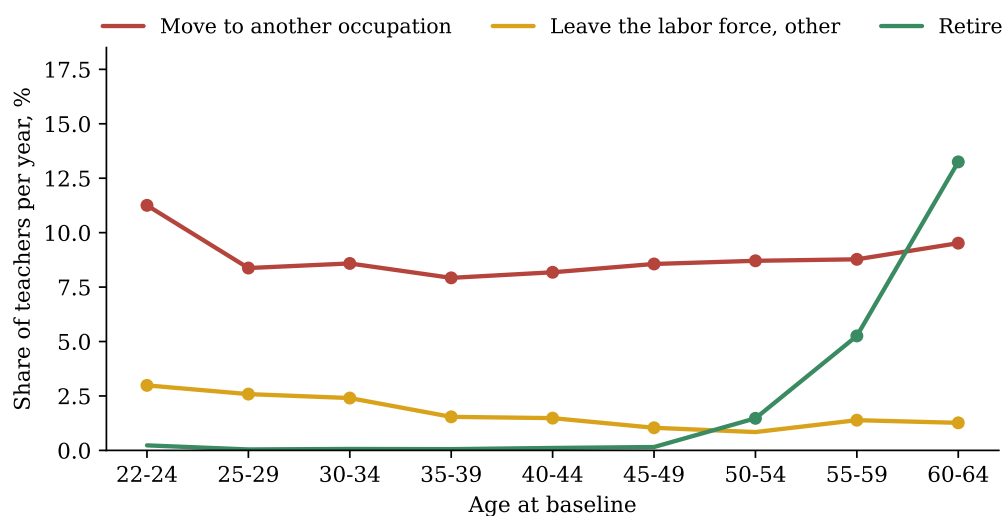


Notes: Weighted composition of non-returning leavers twelve months after teaching, pooling all 2005–2025 waves; the second bar splits the employed by destination occupation. Education jobs include school administration, tutoring and other instruction, teaching assistance, postsecondary teaching, librarianship and other education occupations; care and children occupations include counselors, childcare workers, coaches, social workers, nurses, speech-language pathologists and clergy. Labor force status comes from the second interview. Appendix Figure A1 reports the detailed occupations. Source: author’s calculations from linked CPS pairs, 2005–2025.

The rise of Section 4 has a single engine. It is almost entirely a rise in occupation switching, from 7.3 percent of teachers per year in 2005–2010 to 10.7 percent in 2022–2024, while exits into non-participation have stayed flat at about 4 percent and unemployment spikes only in recessions (Appendix Figure A2). Teaching is increasingly losing people to other jobs, not to inactivity. Mobility inside the profession is substantial as well, since among teachers still teaching a year later 6.7 percent have moved from a public to a private school and 5.7 percent in the opposite direction.

Age reorganizes the same routes completely (Figure 5). Moving to another occupation is not a young teacher’s phenomenon, it is the dominant route at every age below sixty, holding remarkably flat at 8 to 9 percent of teachers per year from the late twenties through the fifties. What changes over the lifecycle is everything else. Family and other labor-force exits fade with age, from 3.0 percent of teachers at ages 22 to 24 to about 1 percent in the late forties, while retirement is invisible before fifty and then takes over, reaching 13.2 percent per year at ages 60 to 64. A district worried about mid-career losses is worried, whether it knows it or not, about competition from other employers.

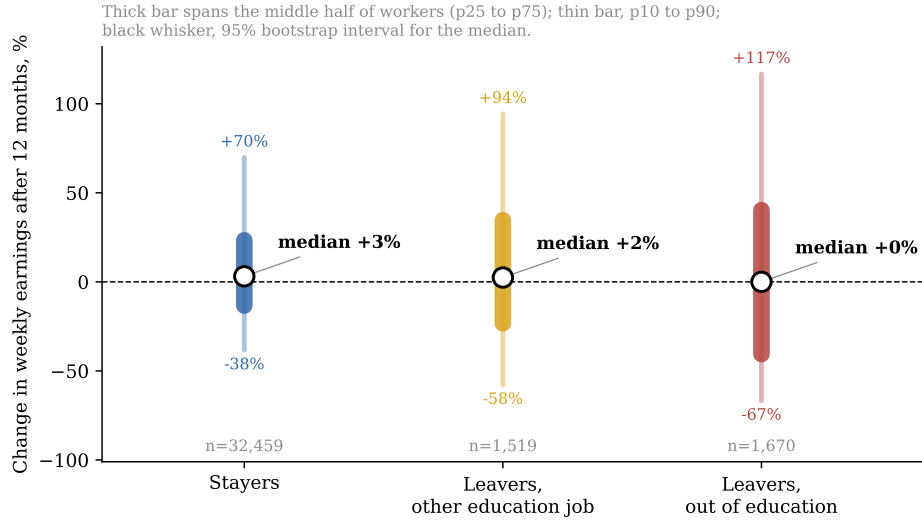
Figure 5: Exit routes over the lifecycle.



Notes: Weighted share of teachers leaving through each route within twelve months, by age at baseline, pooling all 2005–2025 waves under the non-returning definition. Routes come from the labor force status recorded twelve months later; the labor-force route excludes retirement, shown separately, and includes disability. The unemployment route never reaches one percent of teachers at any age and is omitted. Markers are drawn only where a route reaches one percent of teachers per year. Source: author’s calculations from linked CPS pairs, 2005–2025.

Leaving does not pay, at least not on average and not soon. On the subsample with weekly earnings observed a year apart, the median leaver who moves out of education earns no more than she earned teaching, while the median stayer gains 3 percent, and what leaving changes is the risk, since the middle half of earnings changes runs from -41 to $+40$ percent for these leavers against -13 to $+23$ percent for stayers (Figure 6). Moves into other education jobs sit in between on both counts. The absence of a premium at the median, next to the heavy lower tail, suggests that many job-to-job exits are pushed by the classroom rather than pulled by pay, consistent with the modest retention effect of wages in Section 6.

Figure 6: The earnings gamble of leaving.



Notes: Distribution of the change in weekly earnings between the two outgoing-rotation interviews, taken twelve months apart, for teachers with earnings of at least \$50 in both; these baselines have no re-interview window, so leavers follow the conventional 12-month definition, and leavers out of education hold no education occupation at $t+12$. Earnings are collected only in rotations 4 and 8, which is why the samples are smaller than in the rest of the paper. Source: author's calculations from linked CPS pairs, 2005–2025.

6. Who leaves, and why

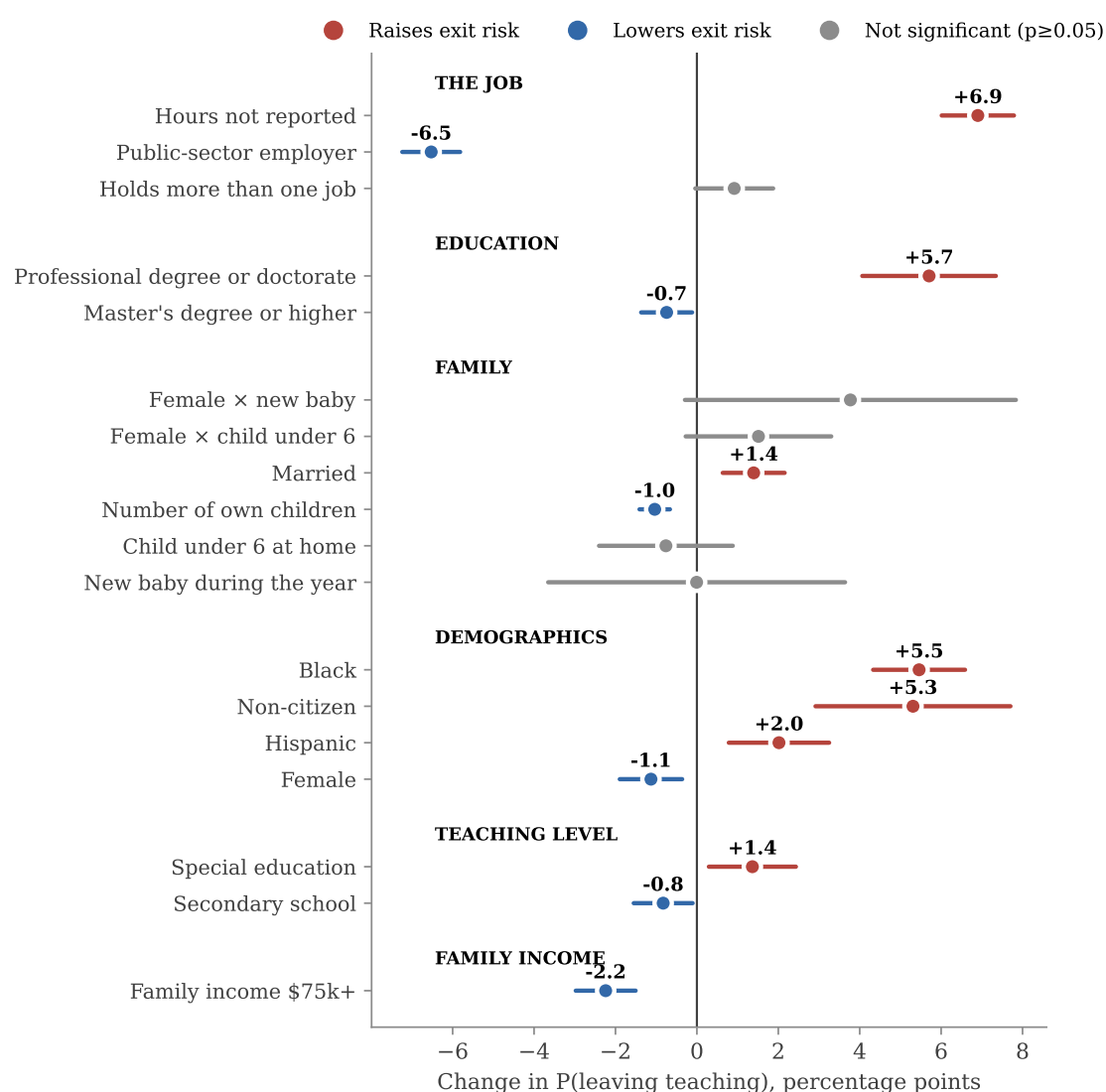
Leaving concentrates among the loosely attached, and the raw contrasts say so before any model. Compared with stayers (Appendix Table A5), leavers are three years older, private-sector employment is eleven points more common among them, they are half again as likely to hold a professional degree or doctorate, and they have fewer young children at home rather than more, while the gender split is economically irrelevant, 75.0 percent of stayers are women against 73.7 of leavers.

To weigh every characteristic at once, I estimate a probit for the probability that teacher i , observed in year t , leaves without an observed return,

$$\Pr(\text{leave}_i = 1 \mid x_i) = \Phi(x_i' \beta + \gamma_t), \quad (1)$$

where x_i collects demographics, family structure, credentials, hours, sector and region at baseline and γ_t are base-year fixed effects. I report average marginal effects in percentage points, with standard errors clustered by household; these are conditional associations, not causal effects. Figure 7 plots the estimates and Appendix Table A8 reports the full model.

Figure 7: Average marginal effects on the probability of leaving teaching, with 95% confidence intervals.



Notes: Estimates from the probit of Table A8. The age profile is shown in Appendix Figure A3. Region indicators and the hours-not-reported indicator are included in the model but not displayed.

The job margin dominates. A private-school teacher is 6.5 percentage points more likely to leave than an otherwise identical public-school colleague (Appendix Table A8, column 2, significant at 1 percent), half of the baseline exit rate of 13 percent and the largest effect in the model. Pay retains, modestly. On the outgoing-rotation subsample of 38,195 teachers with observed earnings, ten percent higher weekly pay lowers the probability of leaving by 0.26 percentage points, about two percent of the baseline rate.

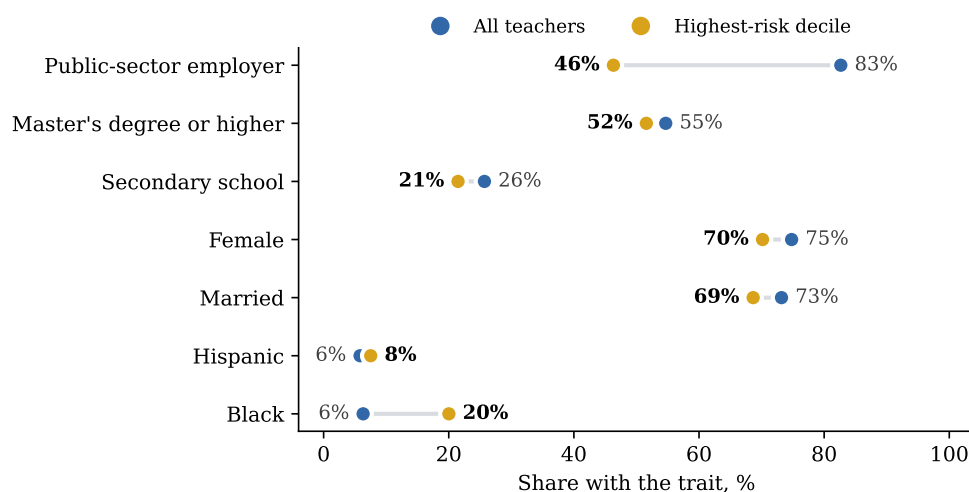
The family results separate stocks from events. Each own child at home predicts about one point less exit (significant at 1 percent), while the event of a birth works in the opposite direction and only for women. A new baby leaves men's exit probability unchanged, a precisely estimated zero with a p-value of 1.0, but raises women's by 3.8 points, marginally significant with a p-value of 0.07, and 56 percent of the new mothers who leave exit the labor force entirely against 30 percent of the average leaver. Black teachers leave at 5.5 points above comparable white teachers and non-citizens at 5.3 points above citizens, both significant at 1 percent and each about 40 percent of the baseline rate, gaps that compound into the diversity of the future workforce. Credentials cut both ways, since a master's retains modestly while a

professional degree or doctorate predicts 5.7 points more exit, consistent with better outside options, and the predicted probability of leaving follows a U in age, lowest in the late thirties and rising steeply after 55 (Appendix Figure A3).

None of this depends on the estimator or the period. A logit, a linear probability model, or dropping the pandemic years 2019–2021 leaves every effect essentially unchanged (Appendix Table A7), and the same model on subsamples shows the job effects are universal, for women and men, public and private, younger and older (Appendix Figure A4). The routes tell more than the total. Estimating the model separately for each exit route, the national counterpart of [Stinebrickner \(2002\)](#) and [Dolton and van der Klaauw \(1999\)](#), splits the predictors into two clean families (Appendix Table A6). The opportunity variables work almost entirely through job switching, since the public-sector advantage is 5.5 of its 6.5 points on the occupation margin and a professional degree or doctorate raises switching by 5.7 points while leaving labor-force exit untouched. The family variables work almost entirely through non-participation, since a new baby moves mothers out of the labor force by 5.2 points while reducing their job switching by 2.3. Pay retains on both margins.

Ranking teachers by their predicted probability of leaving draws the profile of the typical leaver (Figure 8). The top decile of predicted risk faces a mean exit probability of 29 percent, 2.3 times the average teacher, and it is late-career, with a mean age of 57 against 44 for all teachers, far less anchored in the public sector, where 46 percent of the group works against 83 percent of all teachers, and disproportionately Black, 20 against 6 percent, while women are if anything under-represented. Risk accumulates on the attachment margins and at the career extremes, not in any demographic majority.

Figure 8: Characteristics of the top predicted-risk decile vs all teachers.



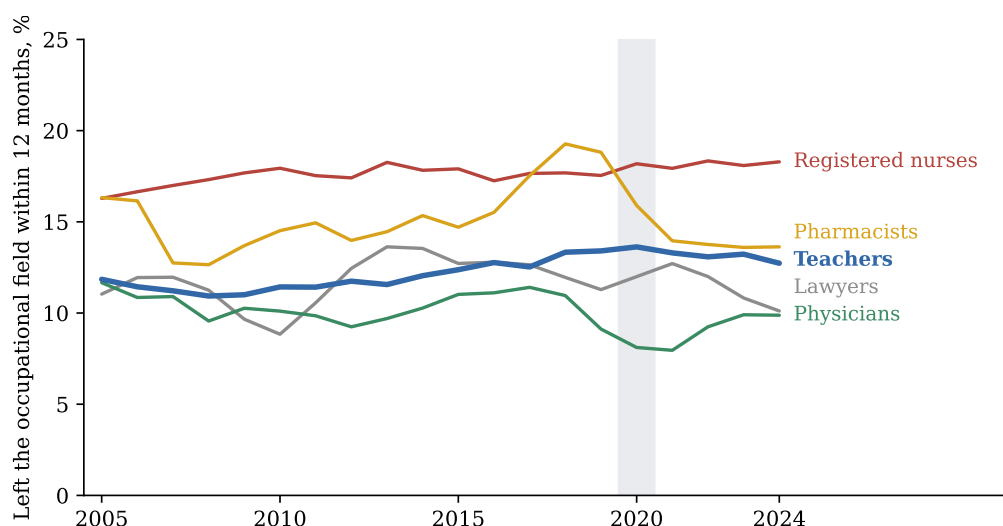
Notes: Predicted probabilities from the probit of Table A8. Source: author's calculations from linked CPS pairs, 2005–2025.

7. Teaching among the professions

One in seven means little until it is judged against how fast similar people leave similar careers, so I build the identical linked panels for seven other degree-holding professions, registered nurses, pharmacists, physical therapists, physicians, lawyers, social workers and accountants. On the amount, teaching is unexceptional. Figure 9 plots exits from the whole occupational field for the five professions with dis-

tinctive, cleanly coded titles, counting moves inside a field, a teacher promoted into school administration or a staff nurse becoming a nurse practitioner, as staying, and teachers leave teaching more slowly than nurses leave nursing or pharmacists leave pharmacy in almost every year, faster only than physicians and lawyers, whose training costs most. No profession in the figure shows a post-pandemic break. Whatever problem teaching has, it is not an exodus, the same conclusion [Harris and Adams \(2007\)](#) and [Aldeman and Yi \(2025\)](#) reach from the retrospective March question; Appendix Table A4 reports my replication of their design next to the linked rates for all seven professions.

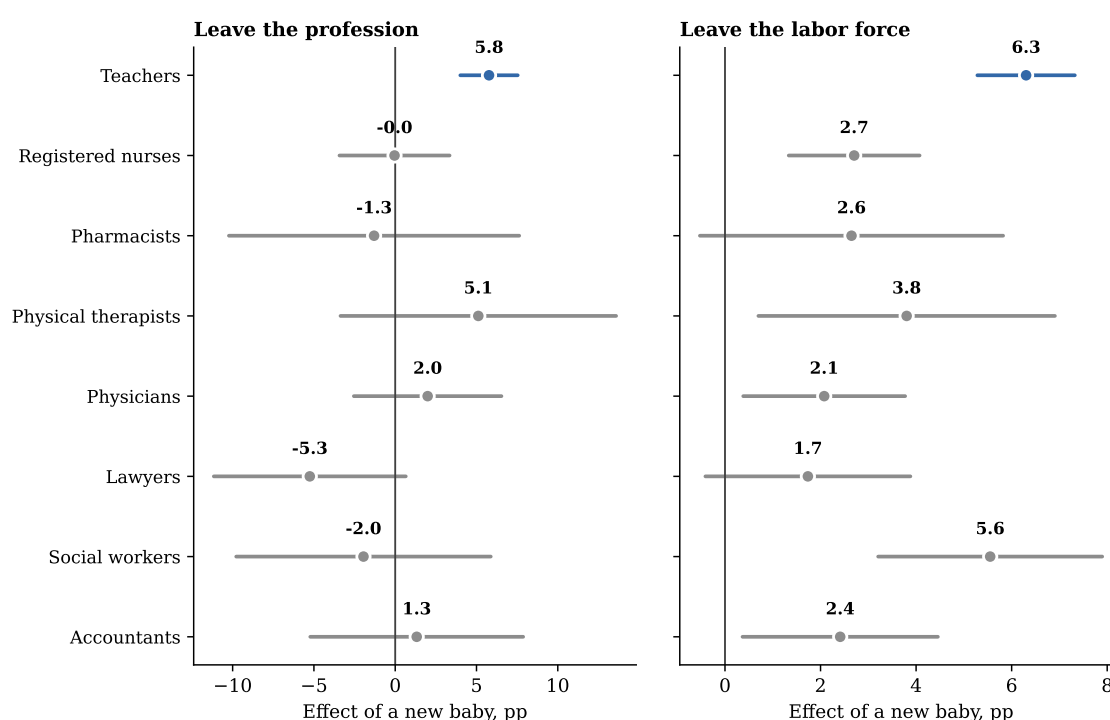
Figure 9: Leaving the occupational field within twelve months, five professions under the same linked design.



Notes: Weighted shares of employed college graduates no longer employed in any occupation of their field twelve months later, education occupations for teachers, health-practitioner occupations for nurses, pharmacists and physicians, and legal occupations for lawyers, linked and validated exactly as the teacher panel and smoothed with a centered three-year moving average. Linked observations are 148,390 teachers, 59,669 nurses, 41,717 lawyers, 30,132 physicians and 8,490 pharmacists. The shaded band marks 2020. Appendix Table A4 reports all seven comparison professions under both the linked and the retrospective instruments. Source: author's calculations from linked CPS pairs, 2005–2025.

The exception appears on the family margin. Section 6 showed that a birth pushes mothers, and only mothers, out of teaching, but that fact alone cannot say whether the push is about teaching or simply about motherhood, since new mothers step back from work in every occupation. To separate the two, Figure 10 runs the identical birth event through every profession, the profession-level counterpart of the child penalty literature ([Kleven, Landais and Sogaard, 2019](#); [Gulek, 2025](#)), and the push turns out to be about teaching. A new baby raises a teacher's probability of leaving her profession by 5.8 percentage points (standard error 0.9), the only significantly positive effect among the eight professions. The mechanism is in the second panel. New mothers pause work everywhere, in nursing by 2.7 points, but the other professions absorb the pause without losing the woman from the field, while teaching converts nearly its entire labor-force exit of 6.3 points into a loss for the profession. Nursing, where shift work makes hours genuinely divisible, loses none of its new mothers, the same part-time mechanism that [Goldin and Katz \(2016\)](#) document for pharmacy, so the profession most often described as family friendly is the one a birth is most likely to end a career in. Fathers show no effect anywhere, in teaching or outside it.

Figure 10: A new baby and the probability that a woman leaves her profession.



Notes: Average marginal effects of a new baby, an own child under three appearing between the two interviews, from probits estimated on women of each profession with age, age squared, marital status, an advanced-degree indicator and base-year fixed effects, with 95% confidence intervals and standard errors clustered by household. Leaving the profession is leaving the whole occupational field under the conventional 12-month definition, so a move inside education or inside health practice is not an exit; the teacher effect of 5.8 points sits next to the 3.8 of the non-returning interaction in Appendix Table A8. Births observed among women range from 132 for physical therapists to 2,734 for teachers. Source: author's calculations from linked CPS pairs, 2005–2025.

8. What it means for policy

The results locate attrition in weak attachment to the job, in private-sector positions, lower pay and both career extremes, rather than in broad demographics, with a rising trend made entirely of moves to other occupations; the teacher at highest predicted risk is late-career and outside the public sector, not the young idealist of the burnout narrative. Four levers follow directly. The wage result implies that pay raises retain, though modestly. The private sector is where teaching careers break. The Black, Hispanic and non-citizen retention gaps deserve monitoring for the diversity of the future workforce. And with one in six leavers back within three months and four in ten of the rest still working in education or care, cheap re-entry pathways such as fast-track rehiring and return bonuses can recover part of the outflow at low cost, because most leavers never really left the field. The comparison professions add the sharpest lever of all, since nursing shows that hours genuinely divisible around a birth keep new mothers in the field, exactly the margin where teaching loses most.

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Appendix A. Sample, representativeness and reconciliation

Table A1 traces how the CPS universe narrows down to my analytic sample. In an average month the survey interviews close to 95,000 adults who represent 241 million people, of whom about 1,570 are full-time school teachers with at least a bachelor’s degree, representing 3.9 million teachers. Only the half of the sample that is in its first rotation block can be linked one year ahead, and the linkage succeeds for 78 percent of teachers. Reassuringly, it succeeds less often for everyone else, 74 percent among other college-educated workers and 71 percent among all employed, a ranking that holds in every single year, so sample losses do not select against teachers. Losses arise because the CPS samples addresses rather than people, which means movers drop out, and since moving is correlated with changing jobs my attrition rates are conservative on this margin. Linked persons keep their population weights, so all weighted rates remain representative, and Table A2 confirms that the linked panel matches the full teacher workforce on every observable.

Table A1: From the CPS universe to the analytic sample.

	Persons	Population represented
Adults interviewed in an average month	94,630	240.6M
employed	57,572	148.1M
school teachers (occ. 2310–2330)	1,834	4.6M
with bachelor’s degree or higher	1,710	4.3M
and working full time	1,573	3.9M
Linked 12 months later (47,633 teachers, 20 waves)	148,390	the same population
with re-interviews after $t+12$: main sample	110,130	the same 4.6M

Notes: Monthly figures are averages over the 251 available months, with weighted population totals beside the sample counts. Teachers are identified by Census occupation codes 2310 to 2330 in the main job, with a bachelor’s degree or higher and full-time hours. The linked rows require a validated match to the interview twelve months ahead, and the last row keeps the baselines whose rotation allows at least one re-interview after the link. Source: author’s calculations from CPS basic monthly microdata, 2005–2025.

Table A2: Composition of the linked panel vs all teacher interviews.

	All teacher interviews	Linked panel
Age, years	42.8	43.8
Female	74.4%	74.5%
White	86.4%	87.8%
Master’s degree or higher	53.6%	55.4%
Public-sector employer	80.4%	81.8%
Unique persons	86,552	47,633

Notes: Weighted means. The first column pools every monthly interview of an employed, degree-holding school teacher between 2005 and 2025; the second column is the baseline of the linked 12-month panel used throughout the paper. Source: author’s calculations from CPS basic monthly microdata, 2005–2025.

Two measurement caveats apply throughout. First, the observation window is short by design. A person leaves the survey for good at most sixteen months after entering it, so a teacher who returns to the classroom more than a few months after the twelve-month link is still counted as a leaver, and all my attrition rates are upper bounds on permanent exit. Second, occupation in the returning rotation is re-coded independently, which mechanically inflates measured switching (Mellow and Sider, 1983; Kambourov and Manovskii, 2013), so levels should again be read as upper bounds while comparisons across groups and over time, the object of this paper, are far less affected.

How does my 13.1 percent compare with the published estimates? Table A3 collects them, and the pattern is that the level tracks the measurement instrument, not a disagreement about the truth. Three instruments have been used. The retrospective March supplement of the CPS asks each respondent the occupation of her main job one year earlier and, applied to teachers, yields 6 to 8 percent (Harris and Adams, 2007; Aldeman and Yi, 2025). The Teacher Follow-up Survey re-interviews once a sample drawn from school staff rosters and reports 8.0 percent for public and 12.0 percent for private teachers (NCES, 2024). My design links two independent interviews twelve months apart and reads higher, 13.1 percent overall and 11.5 for public teachers.

Table A3: Teacher attrition across data sources and measurement instruments.

	Instrument	Population	Years	Rate (%)
<i>A. This paper: linked CPS panel, two interviews twelve months apart</i>				
Non-returning leaver, all teachers	Linked panel	Full-time K–12, public and private	2005–2025	13.1
public schools only	Linked panel	Full-time public K–12	2005–2025	11.5
private schools only	Linked panel	Full-time private K–12	2005–2025	20.2
Conventional 12-month leaver	Linked panel	Full-time K–12, public and private	2005–2025	15.4
<i>B. Retrospective March CPS: one recall question about the job held a year ago</i>				
This paper’s replication	March CPS	Teachers, bachelor’s or higher	2022–2024	5.8
Aldeman and Yi (2025)	March CPS	All teachers	2015–2024	7.6
Harris and Adams (2007)	March CPS	All teachers	1972–2004	n.r.
<i>C. Roster follow-up survey: one re-interview of a sample drawn from school staff lists</i>				
Teacher Follow-up Survey, public	NCES roster	Public K–12	2021–2022	8.0
private	NCES roster	Private K–12	2021–2022	12.0
public and private	NCES roster	All K–12	2021–2022	8.4
<i>D. State administrative payroll records</i>				
Goldhaber and Theobald (2023)	WA payroll	Public, one state	1984–2021	6–8

Notes: Annual rates in percent. Panel A is computed from the linked panel of this paper; the non-returning rows purge teachers who return to a classroom within three months, the conventional row does not. Panel B rows apply the retrospective March-supplement definition, a teacher by the occupation of the longest job held the previous calendar year and a leaver by the survey-week occupation; my replication pools the 2022–2024 supplements, and Aldeman and Yi (2025) pool 2015–2024. Panel C quotes the 2021–22 Teacher Follow-up Survey (NCES, 2024), whose 8.4 percent all-teacher figure Aldeman and Yi (2025) report. Panel D quotes the Washington state range reported by Goldhaber and Theobald and cited in Aldeman and Yi (2025). “n.r.” marks a rate the source does not report separately. Source: author’s calculations and the cited studies.

The gap between my estimate and the retrospective figures is the instrument, and the replication proves it. Applying the recall question to my own CPS files reproduces the retrospective literature exactly, 5.8 percent for degree-holding teachers, against 13.1 percent from the same population measured with linked interviews. A single recall question compresses job-to-job movement, a bias documented for occupational mobility in the CPS and the PSID (Mellow and Sider, 1983; Kambourov and Manovskii, 2013); the linked panel does not, and it records occupation switching at roughly 12 percentage points per year against the 1.9 the recall question sees, while exits from employment are similar under both. The follow-up survey lies between the two because it observes a real transition, but only one, and from a roster that excludes the short-term substitutes and itinerant staff that a household survey’s self-reported “teacher” can include, while its occupation is recorded once rather than coded independently at two interviews. Every one of these differences pushes my level above the roster and recall figures rather than below them, so I read 13.1 percent as an upper bound on the true separation rate. Crucially, the findings of the paper are comparisons and trends within my own consistently measured data, and none of them depends on where that level sits relative to the benchmarks. The same instrument gap appears in every comparison profession, and for the generic titles of accountants and social workers the linked levels are inflated further by independent

re-coding of detailed occupations, which is why Figure 9 plots only the five professions with distinctive titles.

Table A4: Eight professions under the linked and the retrospective instruments.

	Linked design, 2005–2024			Retrospective, 2022–2024	
	Left detailed occupation	Left occupational field	Observations	Left detailed occupation	Sample
Teachers	15.2	12.1	148,390	6.7	7,909
Registered nurses	20.6	17.6	59,669	7.5	5,176
Pharmacists	17.9	15.2	8,490	10.1	486
Physical therapists	21.5	15.1	6,249	11.5	440
Physicians	14.7	10.1	30,132	8.3	1,440
Lawyers	12.6	11.7	41,717	4.5	1,695
Social workers	40.4	32.7	20,296	15.9	1,170
Accountants and auditors	38.1	31.4	44,786	5.6	2,088

Notes: Annual exit rates in percent, weighted. The linked columns come from 12-month occupation pairs of full-time degree-holding workers, built, linked and validated exactly as the teacher panel, pooled over 2005–2024 baselines, under the conventional definition; field exit counts only moves outside the whole occupational field. The retrospective columns replicate the design of Harris and Adams (2007) and Aldeman and Yi (2025) on the 2022–2024 March supplements, comparing the occupation of the longest job held the previous calendar year with the survey-week occupation, without a degree restriction, weighted by supplement weights. Source: author’s calculations from linked CPS pairs and March supplements.

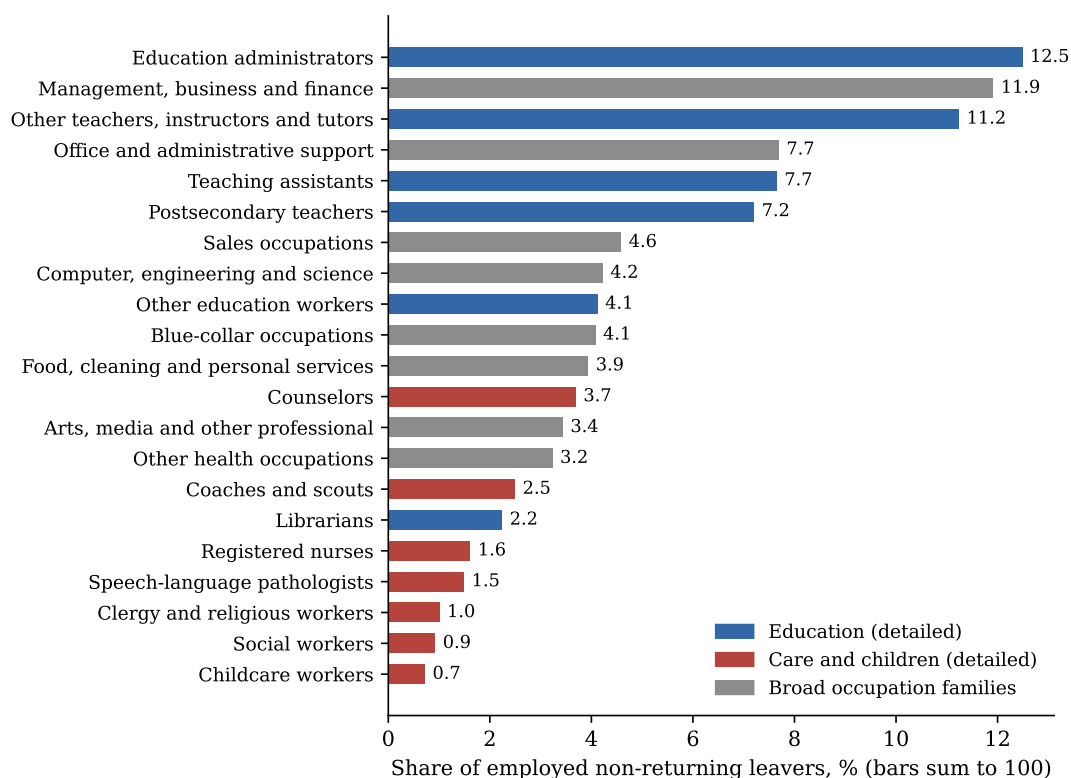
Appendix B. Estimation output

Table A5: Stayers vs non-returning leavers in the analytic sample (weighted means).

	All teachers	Stayers	Non-returning leavers	Difference
Age, years	43.8	43.4	46.6	3.21***
Female	74.5%	74.7%	73.2%	-1.4**
Female aged 25–44	36.9%	37.9%	30.3%	-7.6***
Married	72.0%	72.3%	69.9%	-2.4***
Number of own children (<18) at home	0.8	0.9	0.6	-0.25***
Child under 6 at home	18.1%	18.8%	13.8%	-5.0***
New baby during the year	2.6%	2.6%	2.9%	0.3
Black	8.2%	7.5%	12.3%	4.8***
Hispanic	7.8%	7.6%	8.7%	1.1**
Non-citizen	1.4%	1.3%	2.5%	1.2***
Master's degree or higher	55.4%	55.8%	52.8%	-3.0***
Professional degree or doctorate	2.9%	2.6%	4.8%	2.3***
Holds more than one job	7.8%	7.7%	8.2%	0.5
Public-sector employer	81.9%	83.3%	72.0%	-11.4***
Family income \$75k+	77.6%	78.2%	73.8%	-4.4***
Elementary / middle school	66.7%	66.5%	68.3%	1.9**
Secondary school	25.2%	25.5%	23.5%	-1.9***
Special education	8.1%	8.1%	8.1%	0.0
Observations	110,130	96,099	14,031	

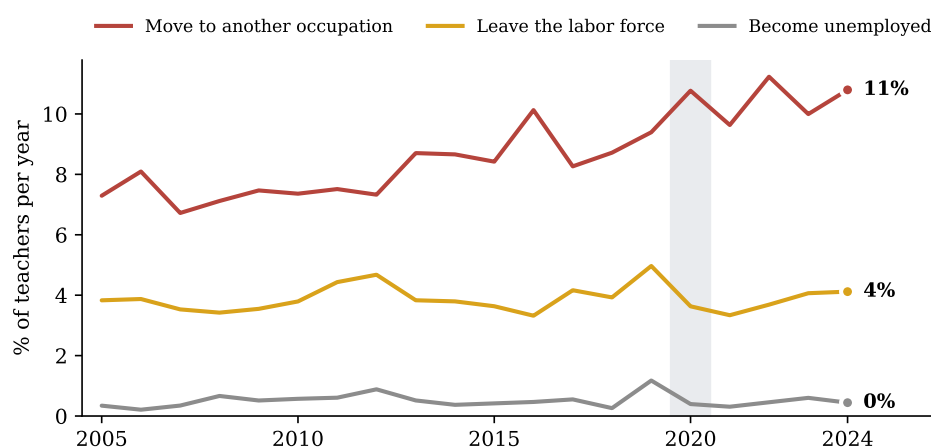
Notes: Weighted means at baseline over the linked analytic sample under the non-returning definition. Shaded rows mark differences that are statistically significant at the 5 percent level and at least two percentage points, or two years, wide. Standard errors are clustered by household. Source: author's calculations from linked CPS pairs, 2005–2025.

Figure A1: Detailed destination occupations of employed non-returning leavers.



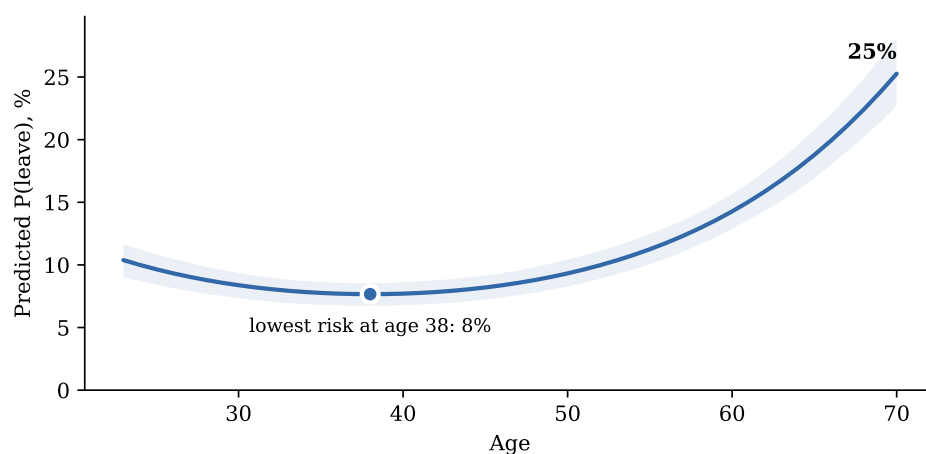
Notes: Weighted shares of non-returning leavers employed twelve months after teaching, 66 percent of all non-returning leavers; bars sum to 100. Education and care occupations are shown at the detailed level and all remaining destinations as broad occupation families. Source: author's calculations from linked CPS pairs, 2005–2025.

Figure A2: Leavers by destination over time.



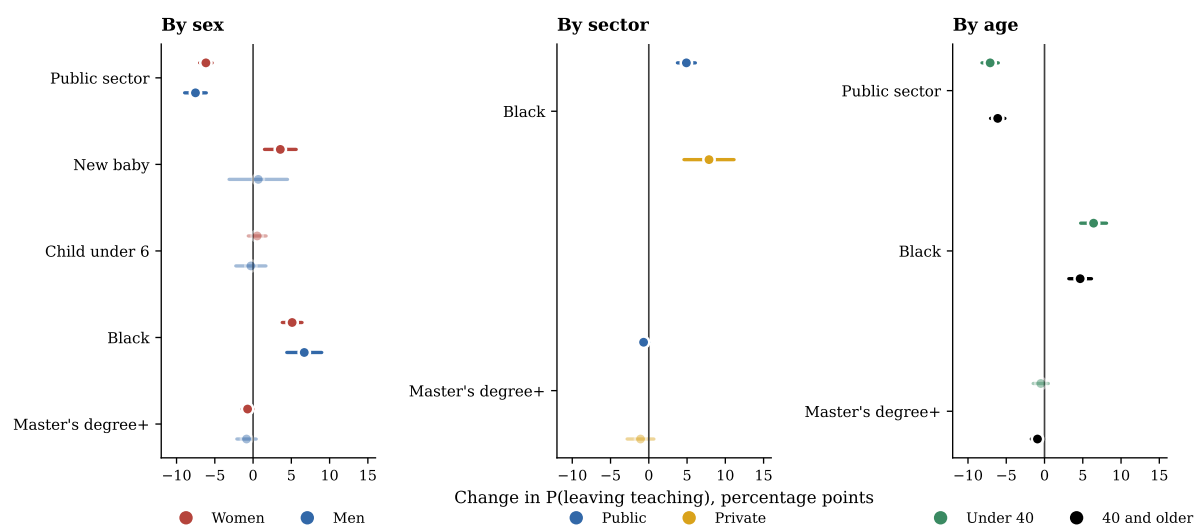
Notes: Non-returning leavers as a share of baseline teachers in each year, split by their situation twelve months later. Source: author's calculations from linked CPS pairs, 2005–2025.

Figure A3: Predicted probability of leaving by age.



Notes: Predicted probabilities from the probit of Table A8, with all covariates other than age and its square at their sample values. Source: author's calculations from linked CPS pairs, 2005–2025.

Figure A4: The same model estimated on subsamples.



Notes: Average marginal effects of the probit of Table A8 re-estimated on each subsample, with 95% confidence intervals and standard errors clustered by household. Source: author's calculations from linked CPS pairs, 2005–2025.

Table A6: Exit routes. Probit average marginal effects by destination.

	(1) Any exit	(2) Another occupation	(3) Out of the labor force	(4) Unemployed
Public-sector employer	-6.53*** (0.36)	-5.53*** (0.29)	-0.54*** (0.20)	-0.25*** (0.06)
Professional degree or doctorate	5.70*** (0.84)	5.74*** (0.65)	-0.59 (0.49)	0.10 (0.15)
Master's degree or higher	-0.74** (0.32)	-0.43 (0.27)	-0.30* (0.17)	-0.14** (0.05)
Black	5.46*** (0.58)	3.95*** (0.47)	1.02*** (0.31)	0.27*** (0.09)
Non-citizen	5.31*** (1.22)	2.88*** (0.97)	1.87*** (0.59)	0.37** (0.17)
New baby during the year	-0.01 (1.86)	-0.20 (1.48)	-0.23 (1.35)	-0.09 (0.34)
Female $\times$ new baby	3.77* (2.07)	-2.25 (1.74)	5.23*** (1.42)	-0.08 (0.39)
Number of own children	-1.04*** (0.19)	-0.44*** (0.15)	-0.88*** (0.12)	-0.12*** (0.04)
Female	-1.13*** (0.39)	-1.56*** (0.32)	0.68*** (0.22)	-0.18*** (0.06)
<i>Outgoing-rotation subsample with weekly earnings</i>				
Log weekly earnings	-2.60*** (0.37)	-1.18*** (0.32)	-0.81*** (0.20)	-0.30*** (0.09)
Base-year fixed effects	Yes	Yes	Yes	Yes
Full covariate set	Yes	Yes	Yes	Yes
Mean of dependent variable	0.127	0.084	0.039	0.005
Observations	110,130	110,130	110,130	110,130
wage subsample	38,195	38,195	38,195	38,195

Notes: Average marginal effects in percentage points from separate probits for any non-returning exit and for each destination, with the full covariate set and base-year fixed effects, standard errors clustered by household in parentheses. The wage row adds log weekly earnings on the outgoing-rotation subsample, where the conventional 12-month definition applies because those baselines have no re-interview window. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Source: author's calculations from linked CPS pairs, 2005–2025.

Table A7: Robustness to estimator and sample. Dependent variable is the non-returning leaver.

	(1) Probit (baseline)	(2) Logit	(3) Linear probability model	(4) Probit, excl. 2019–2021
Public-sector employer	-6.53*** (0.36)	-6.39*** (0.35)	-7.40*** (0.46)	-6.61*** (0.39)
Professional degree or doctorate	5.70*** (0.84)	5.41*** (0.80)	6.98*** (1.15)	5.35*** (0.90)
Master’s degree or higher	-0.74** (0.32)	-0.79** (0.32)	-0.65** (0.32)	-0.76** (0.34)
Black	5.46*** (0.58)	5.30*** (0.56)	6.39*** (0.75)	5.35*** (0.61)
Non-citizen	5.31*** (1.22)	5.17*** (1.15)	6.73*** (1.70)	4.85*** (1.30)
New baby during the year	-0.01 (1.86)	-0.10 (1.93)	0.08 (1.69)	0.23 (1.95)
Female × new baby	3.77* (2.07)	3.89* (2.12)	3.83* (1.99)	3.62* (2.16)
Number of own children	-1.04*** (0.19)	-1.17*** (0.20)	-0.70*** (0.17)	-0.93*** (0.20)
Family income \$75k+	-2.24*** (0.38)	-2.21*** (0.37)	-2.37*** (0.41)	-2.36*** (0.39)
Female	-1.13*** (0.39)	-1.13*** (0.39)	-1.22*** (0.42)	-1.59*** (0.41)
Base-year fixed effects	Yes	Yes	Yes	Yes
Full covariate set	Yes	Yes	Yes	Yes
Mean of dependent variable	0.127	0.127	0.127	0.125
Observations	110,130	110,130	110,130	96,122
Unique teachers	45,109	45,109	45,109	39,450

Notes: Average marginal effects in percentage points, standard errors clustered by household in parentheses. The last column drops the 2019–2021 baselines. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Source: author’s calculations from linked CPS pairs, 2005–2025.

Table A8: Determinants of leaving teaching. Probit, dependent variable is the non-returning leaver.

	Leaves teaching with no observed return	
	(1) Probit coefficients	(2) Marginal effect (percentage points)
Age (years)	-0.057*** (0.005)	-1.12*** (0.09)
Age ² /100	0.074*** (0.005)	1.47*** (0.10)
Female	-0.057*** (0.020)	-1.13*** (0.39)
Married	0.071*** (0.020)	1.40*** (0.39)
Number of own children	-0.052*** (0.010)	-1.04*** (0.19)
Child under 6 at home	-0.038 (0.042)	-0.76 (0.84)
Female × child under 6	0.076* (0.046)	1.51* (0.91)
New baby during the year	-0.000 (0.094)	-0.01 (1.86)
Female × new baby	0.191* (0.105)	3.77* (2.07)
Black	0.276*** (0.029)	5.46*** (0.58)
Hispanic	0.102*** (0.032)	2.02*** (0.63)
Non-citizen	0.268*** (0.062)	5.31*** (1.22)
Master's degree or higher	-0.038** (0.016)	-0.74** (0.32)
Professional degree or doctorate	0.288*** (0.042)	5.70*** (0.84)
Hours not reported	0.349*** (0.023)	6.90*** (0.45)
Holds more than one job	0.046* (0.025)	0.92* (0.49)
Public-sector employer	-0.330*** (0.018)	-6.53*** (0.36)
Family income \$75k+	-0.113*** (0.019)	-2.24*** (0.38)
Midwest	-0.088*** (0.024)	-1.74*** (0.47)
South	0.024 (0.022)	0.47 (0.43)
West	-0.014 (0.024)	-0.28 (0.47)
Secondary school	-0.042** (0.019)	-0.83** (0.37)
Special education	0.069** (0.028)	1.36** (0.55)
Base-year fixed effects	Yes	Yes
Mean of dependent variable	0.127	0.127
Observations	110,130	110,130
Unique teachers	45,109	45,109
Pseudo R^2	0.050	

Notes: Column 1 reports probit coefficients and column 2 average marginal effects in percentage points, with standard errors clustered by household in parentheses. Base-year fixed effects included. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Source: author's calculations from linked CPS pairs, 2005–2025.